INCH-POUND

A-A-50180A
March 15, 1990
SUPERSEDING
A-A-50180
October 20, 1988

COMMERCIAL ITEM DESCRIPTION

FRYERS, DEEP FAT, ELECTRIC

The General Services Administration has authorized the use of this commercial item description in preference to type I and type II, sizes 3, 4, and 5 of S-F-695.

1. CLASSIFICATION

1.1 Classification. This commercial item description covers deep fat fryers in the following types, sizes, and styles (see 5.2).

Type I - Counter Model

Size 1 - 27 pounds minimum production capacity

Type II - Floor Model

Size 1 - 60 pounds minimum production capacity
Size 2 - 90 pounds minimum production capacity
Size 3 - 125 pounds minimum production capacity

Style 1 - With automatic filtering system
Style 2 - Without automatic filtering system

1.2 Part identification number (PIN). The part identification number shall consist of the commercial item description number, type number, size letter, and style number as follows:

A-A-50180A - I A 1
Style
Size
Type
CID number

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AMSC N/A FSC 7310

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2. SALIENT CHARACTERISTICS

2.1 Design and construction. Each fryer shall be equipped with a stainless steel cover with handle for covering the cooking vessel and with one large and two half baskets. Baskets shall be made of stainless steel type 304 series of ASTM A167, or steel plated with a material meeting the requirements of National Sanitation Foundation (NSF) Standard No. 4. Automatic basket lifts shall be furnished upon request. Fryers shall be equipped with stainless steel swing-up heating elements or a control panel including heating elements which can be lifted off for cleaning. Fryers equipped with lift-up heating elements shall be provided with a device to hold or lock the elements in the raised position. The type I fryer shall be designed for counter use and comply with the requirements of table I. The cabinet front door and sides shall be made of type 304 stainless steel of ASTM A167. The back and bottom shall be stainless steel, galvanized steel, or steel treated to resist corrosion. The type I cabinet shall be equipped with 4-inch high adjustable steel legs. The type II cabinet shall be equipped with the option of either 6-inch adjustable stainless steel legs or two rigid and two swivel casters with brakes. Cooking vessels should be stainless steel type 300 series of ASTM A167 or corrosion-resisting steel used for food contact type 400 series of ASTM A176, be removable or non-removable. Non-removable vessels shall be equipped with a hand operated drain valve. A die stamped oil capacity indicating mark shall be provided on the rear wall inside cooking vessel. The cabinet shall be enclosed and have a door or drawer and have a shelf or other means at the bottom to accommodate a fat draining vessel. The cabinet front shall be stainless steel with the remaining side and back finished in accordance with the manufacturer's practice. Type II fryers shall be provided with a stainless steel or aluminum fat draining vessel with carrying handles and a minimum capacity of 35 percent of the cooking capacity. The type II fryers shall comply with the requirements of table I.

TABLE I. Fryer requirements

	Production capacity 1/ Lbs/Hr(min)	Preheat time 70°F to 350°F minutes (max)	Total kW (min)	Dimensions, inches (max)		
				Height 2/	Width	Depth
Type I, size 1	27	15	5.5	14	18	24-1/2
Type II, size 1	60	15	12.0	37	22	39
Type II, size 2	90	15	16.0	37	22	39
Type II, size 3	125	15	16.5	37	22	39

1/ Production capacity of raw to done, 1/2 inch by 1/2 inch, french fries that can be processed in 1 hour.

2/ Height to top of cooking vessel including legs.

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2.1.1 Filtering system. The filtering system shall be an automatic device installed in the base of a single or double basket fryer and so located as not to impede the operation of the fryer. Installation of the device, pan, motor, and hoses shall not increase the overall height of the fryer cabinet. The filtering system shall be completely equipped with all necessary components (motor, pan, hoses, switches, etc.) to insure that the system will remove all sediment from the cooking medium when in the operating mode. The system shall operate so that while filtering the oil of one fryer, other fryers will remain operational. As an alternate, the filtering system may be housed in a separate spreader cabinet. When specified (see 5.2), the fryer shall be furnished with a spreader cabinet fabricated of stainless steel and designed to fit between two adjacent fryers or to locate at the end of single line of fryers. The spreader cabinet shall be the same width as the fryer and shall have a front opening door. The spreader cabinet shall be of the same height and depth as adjacent fryer.

2.2 Electrical requirements. Type I fryer shall operate on a nominal 220/240 volt, 60 hertz, 1-phase power source or a nominal 220 volt, 50 hertz, 1-phase power source as specified (see 5.2). The type II fryers shall operate on a nominal 208 volt, 60 hertz, 3-phase power source or a nominal 380 volt, 50 hertz, 3-phase power source as specified (see 5.2). All fryers shall have provisions for permanent electrical connection.

2.2.1 Solid-state. The solid-state control system shall incorporate, as a minimum, a temperature dial, power ON-OFF switch, start/reset switch, test switch, melt/fry switch, and fuse holder and fuse. All switches shall be located on a single control panel located on the front to the fryer. Switches shall be of the toggle, rocker, or membrane type. As each system is energized, the indicator light for that function shall operate as intended by the manufacturer, i.e., glows, flashes, or goes out.

2.3 Thermostats. The fryers shall be equipped with an adjustable thermostat having a temperature range of 200°/275°F to 375°/400°F. The thermostat shall have a positive "OFF" position with the temperature marked at 50°F intervals and with graduated markings at 25°F intervals. A separate "ON-OFF" switch may be provided in addition to or in lieu of a positive "OFF" position to control power to the thermostat. In addition, the fryer shall be equipped with a non-adjustable overtemperature thermostat of the manual reset type. A signal light shall indicate when the overtemperature thermostat is activated. A test circuit with a signal light shall be provided to ensure the operation of the overtemperature circuit.

2.3.1 Signal lights. The fryer shall be equipped with a signal light to indicate that power is available at the fryer when "ON-OFF" switch is in the "ON" position. The fryer shall have a signal light that glows when the heating elements are heating and goes out whenever the dialed temperature is reached. Signal lights shall be of such size and located in such a manner as to be readily visible by the operator.

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2.4 Codes and standards. The fryers shall comply with Underwriters Laboratories Inc. Standard No. 197 and National Sanitation Foundation (NSF) Standard No. 4.

2.5 Data name plate (Government unique requirement, see 5.4). The fryer shall be furnished with a data name plate in accordance with the applicable requirements of MIL-STD-130 except the requirements for: (a) Methods of applying; (b) Identification tags; (c) Information not required and (d) Optional marking information shall not apply. The data plate shall be made of minimum 20 gauge corrosion-resisting metal and attached to the fryer by rivets, screws, or welding in such a manner as to meet the applicable National Sanitation Foundation sanitary requirements for this equipment. The plate shall contain the following information which shall be stamped, engraved or applied by photosensitive means:

National Stock Number
Procurement Contract Number
Specification Data
Manufacturer's Name, Address and Telephone Number
Supplier's Name, Address and Telephone Number (List only if different from Manufacturer)
Manufacturer's Model Number
Government Approved Manual Number (when specified, see 5.2)

Each plate shall be placed so that it is readily visible to the operator during normal operating use and so as to not adversely affect the life and utility of the fryer.

3. QUALITY ASSURANCE

3.1 Certification. The contractor shall certify, and maintain substantiating evidence, that the product offered meets the salient characteristics and requirements of this Commercial Item Description, and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

3.2 NSF. Acceptable evidence of meeting the requirements of NSF shall be one of the following:

(1) A listing in the current edition of the NSF "Listing of Food Service Equipment" and display of the NSF seal on the finished stand and fryer.

(2) A certification for the fryer issued by NSF under their special one-time contract evaluation/certification service.

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(3) A certified test report from a recognized independent testing laboratory acceptable to the Government agency for which the stand and fryers are being procured indicating that the stand and fryers have been tested and conform to NSF Standard No. 4.

3.3 UL. Acceptable evidence of meeting the applicable requirements of UL 197 shall be the UL listing mark, or a certified test report from a recognized independent testing laboratory, acceptable to the Government, indicating that the fryers offered conform to the applicable requirements of UL 197.

3.4 End item examination. The contractor is responsible for assuring that the product offered conforms to the stated requirements. Sampling inspection shall be in accordance with MIL-STD-105. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 4.0. The sample unit shall be one complete end item. The defects are as follows:

Any missing component
Any burr, sliver, sharp edge, rough die, tool, gauge, trim, or grind marks
Not constructed as specified
Not fabricated of materials specified
Any component fractured, bent, buckled, dented, punctured, or malformed
Fastening devices missing
Welding not smooth and free of cracks, burn holes, or undercuts
Signal lights and switches not provided
Manufacturer's identification marking missing or incomplete
Controls not accessible from the front

3.5 End item testing. One fryer from the production lot shall be tested as specified below. Any test failure shall be cause for rejection of the lot.

3.5.1 Frying capacity. The fryer shall be operated in accordance with the manufacturer's standard procedures as described in the manufacturer's operator's manual. Failure of the fryer to produce an acceptable product in the allotted time shall be a test failure.

3.5.2 Signal lights and switches. All switches and signal lights shall be tested for operation. Any non-functioning switch or light shall be a test failure.

3.5.3 Thermostats. Thermostats shall be tested for accuracy within limits established by the manufacturer. Failure of the thermostat to function within these limits shall be a test failure.

3.5.4 Leakage and spillage. The fryer filled to its rated capacity shall be examined for spillage and leakage during its operating cycle. Any spillage or leakage shall be a test failure.

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3.5.5 Filter system. When the fryer is furnished with a filtering system, the manufacturer shall demonstrate that the system produces a sediment free cooking medium without disturbing operation of adjacent fryers. The filtering operation shall not exceed 5 minutes in duration.

3.6 Metric product. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this CID are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

3.7 Regulatory requirements. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practicable.

4. PACKAGING

4.1 Packaging and packing. Packaging and packing shall be commercial or export as specified (see 5.2).

4.2 Commercial packaging and packing. Each fryer shall be cleaned, preserved, packaged, packed, and unitized in accordance with ASTM D 3951.

4.3 Export.

4.3.1 Export packaging. Each fryer shall be disassembled to the extent determined by the contractor to protect parts susceptible to damage during shipment and storage and for reduction of cubage. Parts removed shall be packaged in a domestic fiberboard box constructed and closed in accordance with PPP-B-636. Parts not removed shall be secured in place with cushioning or pressure-sensitive tape. Doors and drawers shall be secured in a closed position with filament reinforced pressure-sensitive tape. The fryer and parts shall be cleaned, preserved, and cushioned to afford protection against corrosion, deterioration, and physical damage during shipment and storage.

4.3.2 Export packing. Each fryer shall be packed in a sung-fitting wood-cleated, plywood box, conforming to domestic type, style A or J, grade B, type 2, load of PPP-B-601 or nailed wood box conforming to class 1, type 2 or 4, grade B, type 2 load of PPP-B-621. The contents of each box shall be blocked and braced to prevent movement while the box is in transit. When legs or

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casters are not dissassembled, the fryer shall be blocked in a manner to provide approximately 1 inch clearance between the legs or casters and the bottom of the box. Each shipping container shall be closed and strapped in accordance with the appendix of PPP-B-601 and PPP-B-621, as applicable.

4.4 Marking (commercial and export). Marking of unit pack and shipping container shall be in accordance with MIL-STD-129.

5. NOTES

5.1 Intended use. The fryers are intended for use in kitchens or galleys for deep fat frying of food products.

5.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. Type, size, and style required (see 1.1).
c. When a spreader cabinet is required (see 2.1).
d. Power source required (see 2.2).
e. Government approved manual number to be included on the data name plate when manuals are purchased under the contract (see 2.5).
f. Selection of packaging and packing required (see 4.1).

5.3 Sources of documents.

5.3.1 Sources of nongoverment association documents.

ASTM D 3951 - Standard Practice for Commercial Packaging, is available from:

The American Society for Testing and Materials (ASTM)
1916 Race Street
Philadelphia, PA 19103

UL Standard No. 197 - Commercial Electric Cooking Appliances, is available from:

Underwriters Laboratories, Inc. (UL)
333 Pfingsten Road
Northbrook, IL 60062

NSF Standard No. 4 - Commercial Cooking and Hot Food Holding Equipment, is available from:

National Sanitation Foundation (NSF)
3475 Plymouth Road
P.O. Box 1468
Ann Arbor, MI 48106

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5.3.2 Source of governmental documents. Copies of Military and Federal documents are available from:

Standardization Documents Order Desk
Bldg. 4D
700 Robbins Avenue
Philadelphia, PA 19111-5094

5.4 Government unique requirements. Whenever "(Government unique requirements)" is included in the title of a paragraph under "salient characteristics" it is meant that the requirement is something that is not normally offered to the commercial marketplace by the manufacturer.

5.5 Classification comparison.

S-F-695G	CID A-A-50180
Type I	Type I
Type II	
Size 2	
Size 3	Size 1
Size 4	Size 2
Size 5	Size 3

MILITARY INTERESTS:

Custodians:

Army - GL
Navy - YD
Air Force - 99

Review activities:

Army - MD, TS
Navy - SA
Air Force - 84, 50
DLA- GS

User activity:

Navy - MC

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA - FSS
VA - OSS

PREPARING ACTIVITY:

Army - GL

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